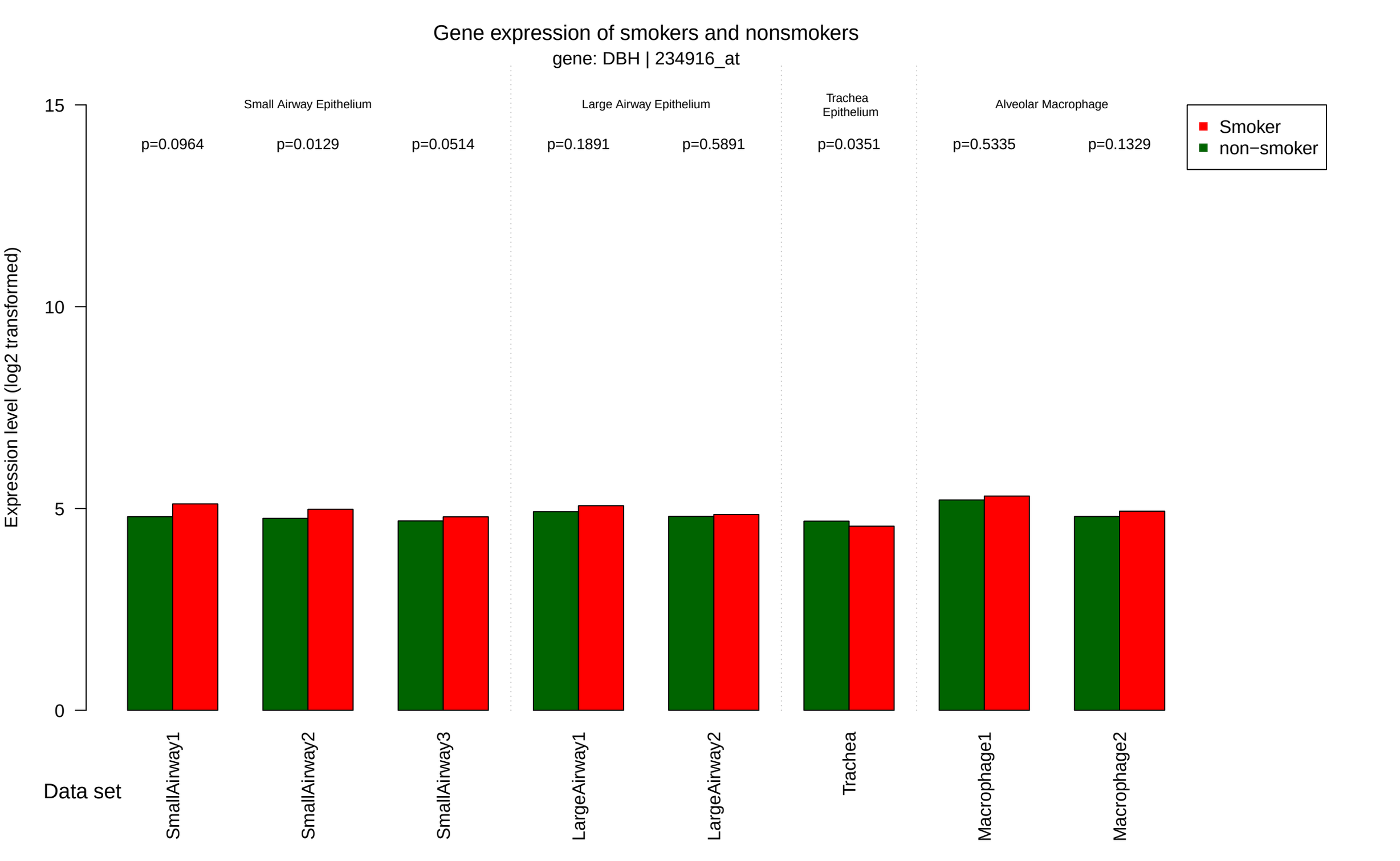
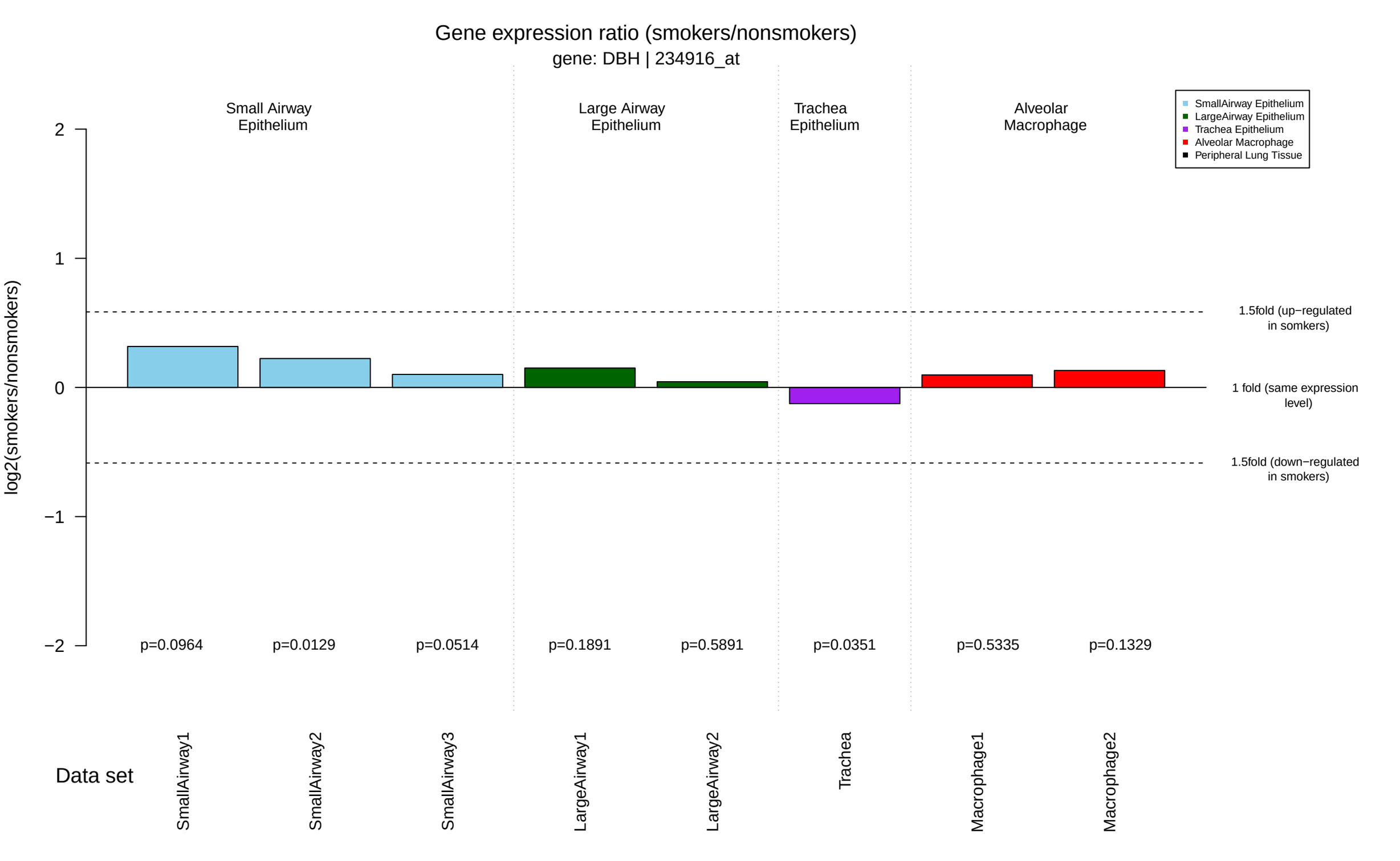
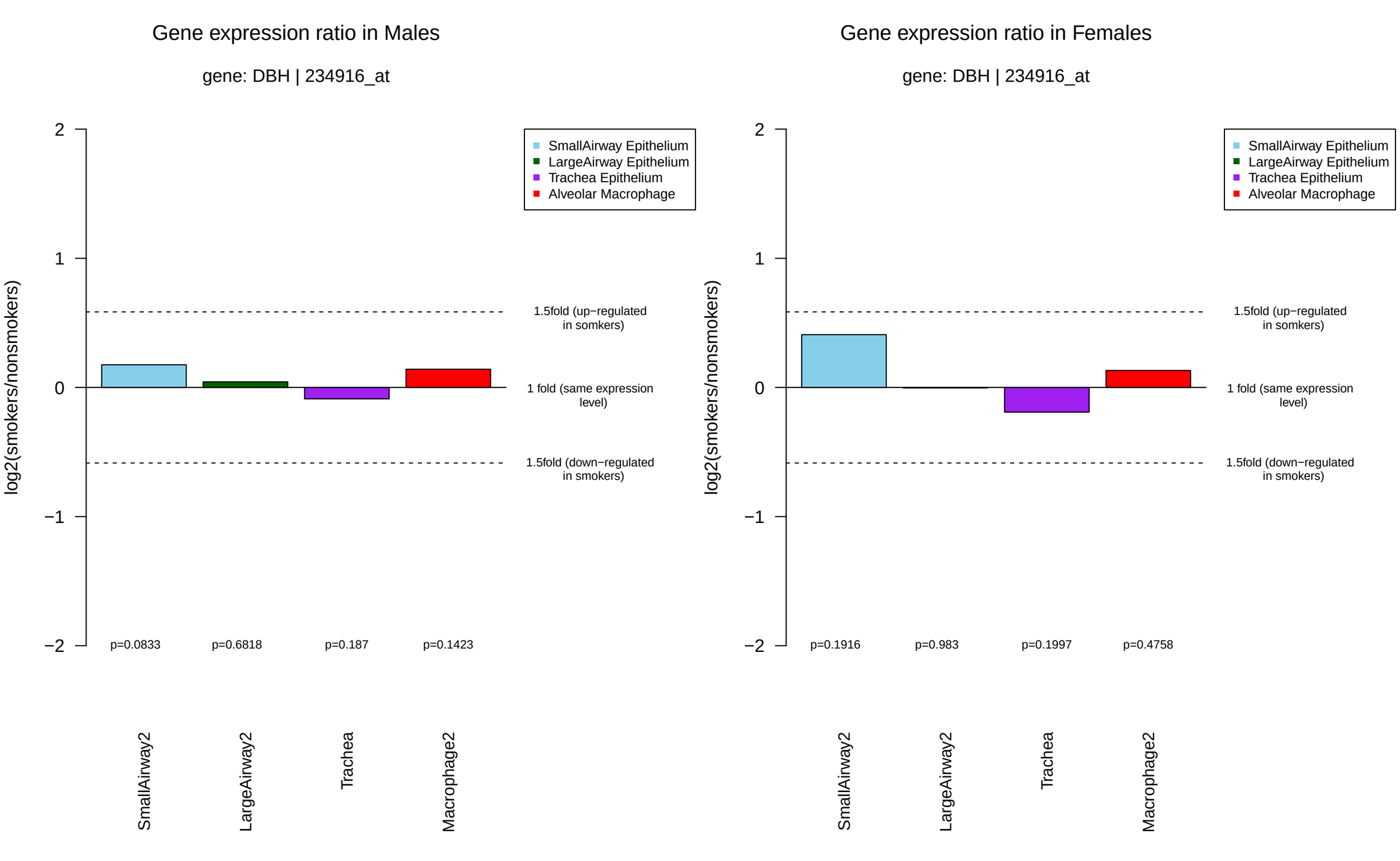
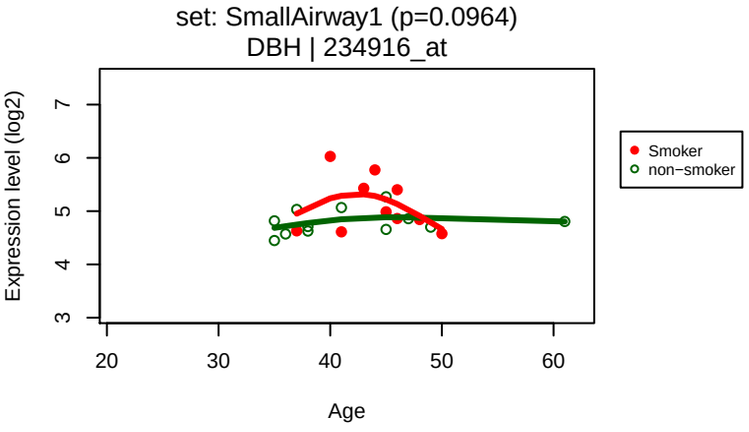
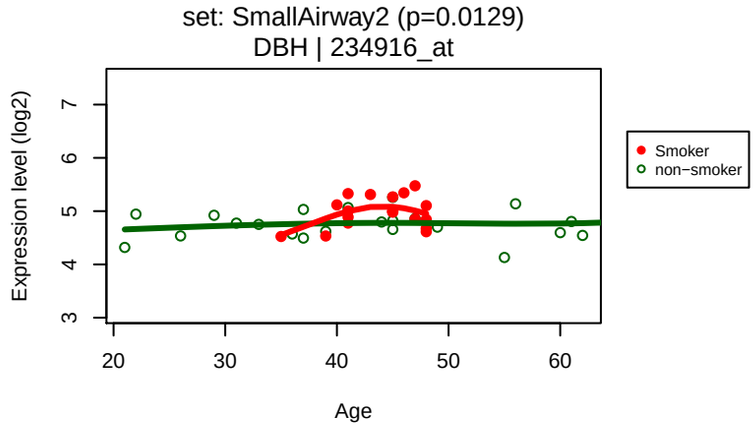
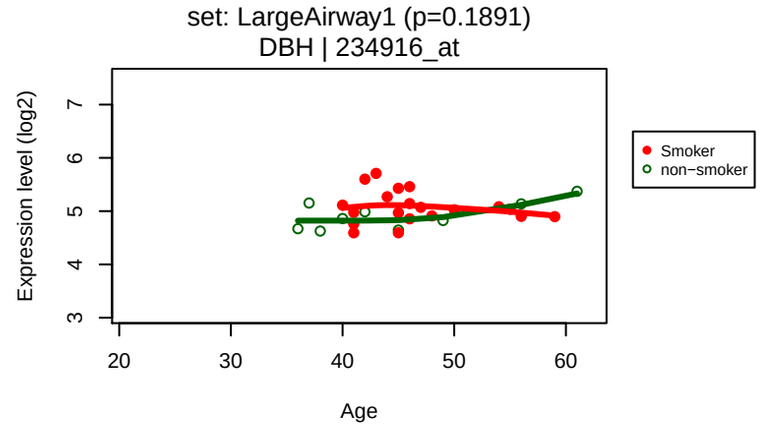

Gene expression of smokers and nonsmokers
gene: DBH | 234916_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0964
p=0.0129
p=0.0514
p=0.1891
p=0.5891
p=0.0351
p=0.5335
p=0.1329
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

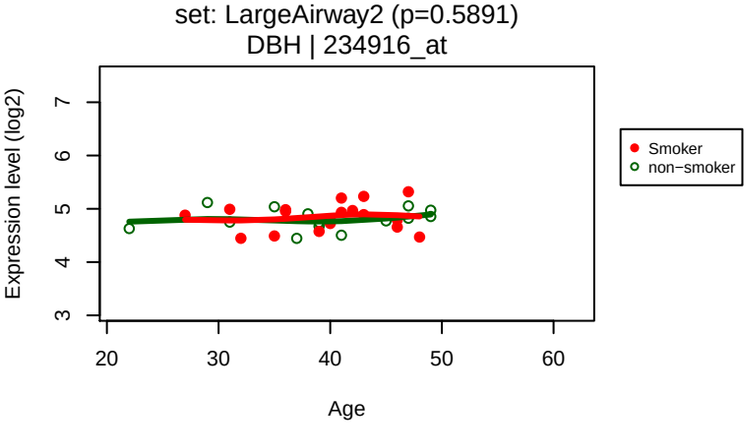

Gene expression ratio (smokers/nonsmokers)
gene: DBH | 234916_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0964
p=0.0129
p=0.0514
p=0.1891
p=0.5891
p=0.0351
p=0.5335
p=0.1329
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

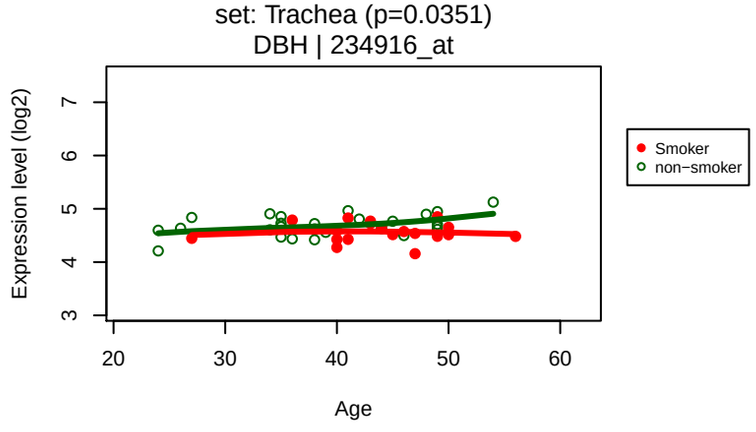

Gene expression ratio in Males
gene: DBH | 234916_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0833
p=0.6818
p=0.187
p=0.1423
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DBH | 234916_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1916
p=0.983
p=0.1997
p=0.4758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

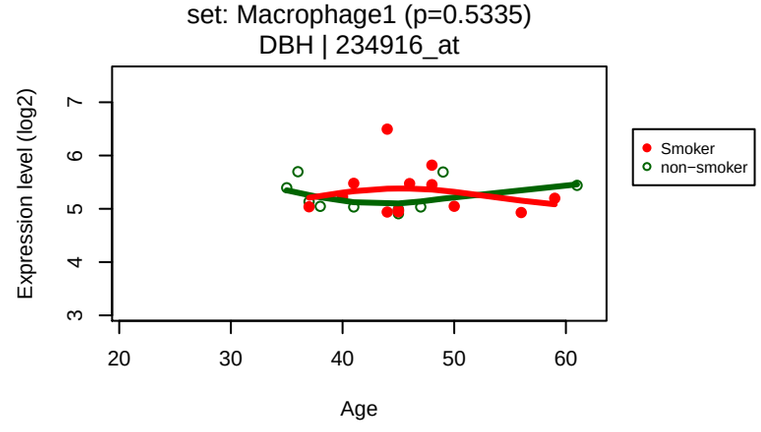

set: SmallAirway1 (p=0.0964)
DBH | 234916_at
Expression level (log2)
Age
Smoker
non−smoker

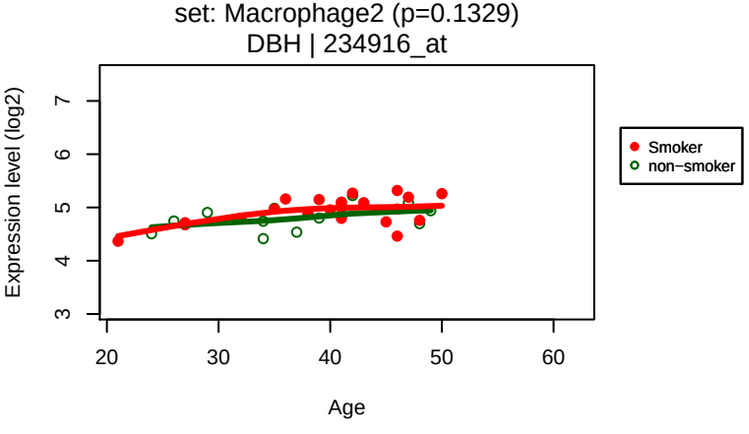

set: SmallAirway2 (p=0.0129)
DBH | 234916_at
Expression level (log2)
Age
Smoker
non−smoker

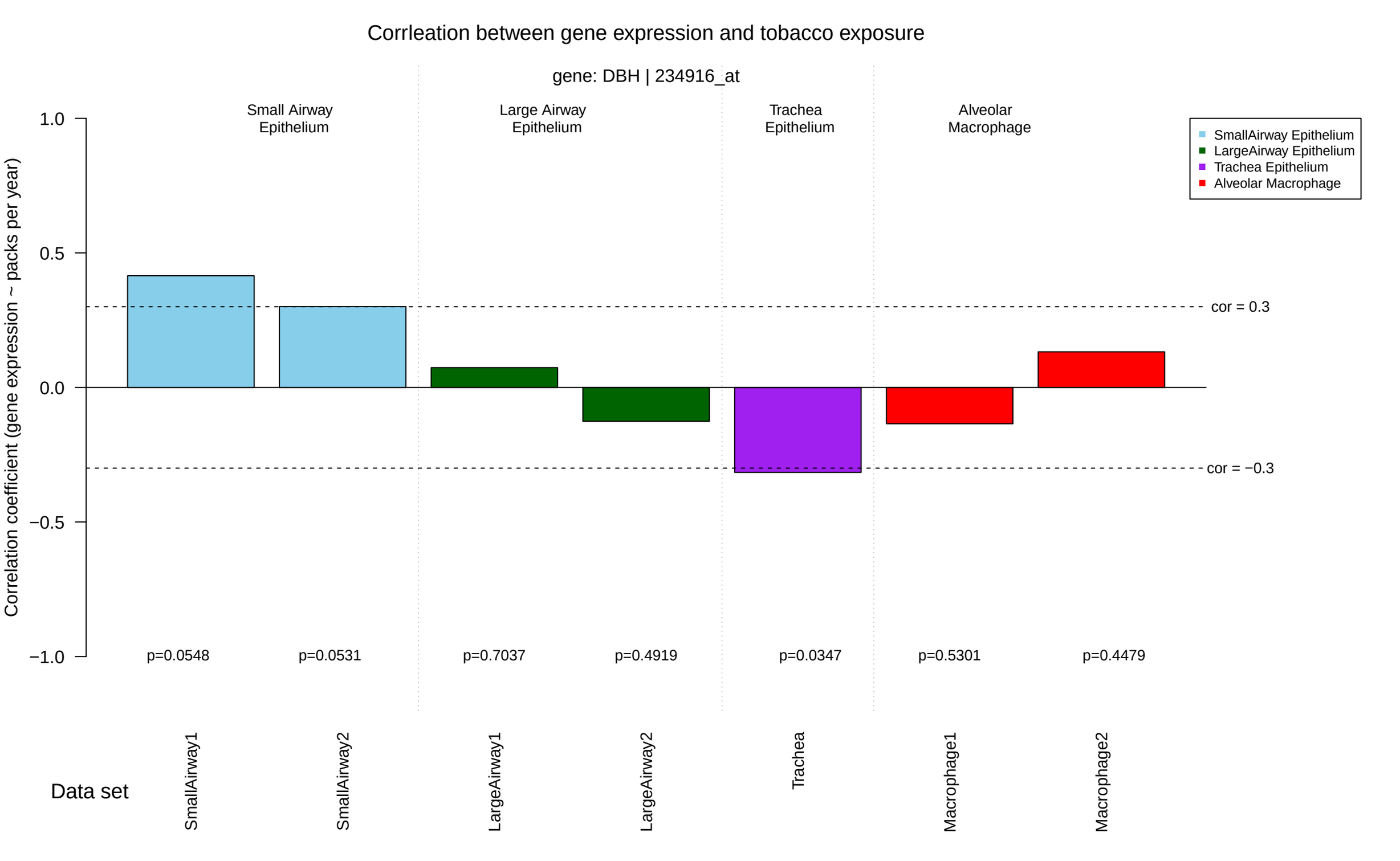

set: LargeAirway1 (p=0.1891)
DBH | 234916_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5891)
DBH | 234916_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0351)
DBH | 234916_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5335)
DBH | 234916_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1329)
DBH | 234916_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DBH | 234916_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0548
p=0.0531
p=0.7037
p=0.4919
p=0.0347
p=0.5301
p=0.4479
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2